

10.14 THE RENORMALIZATION SCALE AND THE RENORMALIZATION GROUP

Two complementary views of the same scale dependence: first the practical one, how the renormalization scale μ is chosen and varied in a calculation; then the conceptual one, in what sense the “renormalization group” is a group.

THE RENORMALIZATION SCALE μ IN THE $\overline{\text{MS}}$ SCHEME

In quantum field theories such as Quantum Chromodynamics (QCD), calculations often encounter infinities from quantum loop corrections in Feynman diagrams. To obtain finite, physical predictions, we employ renormalization, which absorbs these divergences into redefinitions of parameters like masses and coupling constants (e.g., the strong coupling α_s).

The $\overline{\text{MS}}$ (Modified Minimal Subtraction) scheme is a widely used renormalization method (§10.13). It subtracts only the divergent parts (poles in dimensional regularization) and certain constants. In this scheme, we introduce the renormalization scale μ , which at first appears unphysical since physical observables should not depend on our calculational choices. However, μ plays a vital role in capturing how parameters evolve with energy scales.

Consider μ as a “ruler” for measuring interactions at different energy “zoom levels.” In QCD, the strong coupling α_s is not constant but “runs” due to quantum effects, exhibiting asymptotic freedom: it is strong at low energies and weak at high energies. The renormalization group equation (RGE) describes this evolution:

$$\frac{d\alpha_s}{d\ln(\mu^2)} = -\beta(\alpha_s),$$

where, in this convention, $\beta(\alpha_s) = \beta_0\alpha_s^2 + \dots$ with $\beta_0 = (33 - 2n_f)/12\pi > 0$ for n_f quark flavors (note the opposite overall sign from $\beta = \mu dg/d\mu$ in §10.13). At one loop the solution is

$$\alpha_s(\mu) \approx \frac{\alpha_s(\mu_0)}{1 + \beta_0\alpha_s(\mu_0)\ln(\mu^2/\mu_0^2)}.$$

Here, α_s explicitly depends on μ . Measuring α_s at one scale (e.g., $\mu = M_Z \approx 91$ GeV) allows prediction at others via the RGE. This running is physical and experimentally verified.

While μ is a technical parameter, it is dynamical, reflecting the physical scale of the process. In perturbative calculations, terms like $\ln(q^2/\mu^2)$ appear, where q is a momentum. In idealized processes with a single hard scale, such as the center-of-mass energy $\sqrt{s}$ in $e^+e^- \rightarrow \text{hadrons}$, we set $\mu^2 = s$ for several reasons:

1. **Minimizing Logarithms:** This makes $\ln(s/\mu^2) \approx 0$, reducing higher-order corrections and improving perturbation theory reliability.
2. **Resumming Effects:** It incorporates some higher-order contributions into the running coupling.
3. **Physical Matching:** μ separates short- and long-distance physics; aligning it with the dominant scale captures the dynamics effectively.
4. **Smaller Truncation Error:** Good choices minimize the error from truncating the series at finite order.

Choosing μ inappropriately can produce large logarithms, and, if μ is too small, a large α_s , breaking perturbation theory.

HOWEVER, $\mu^2 = s$ IS MERELY A “BEST-GUESS” CHOICE

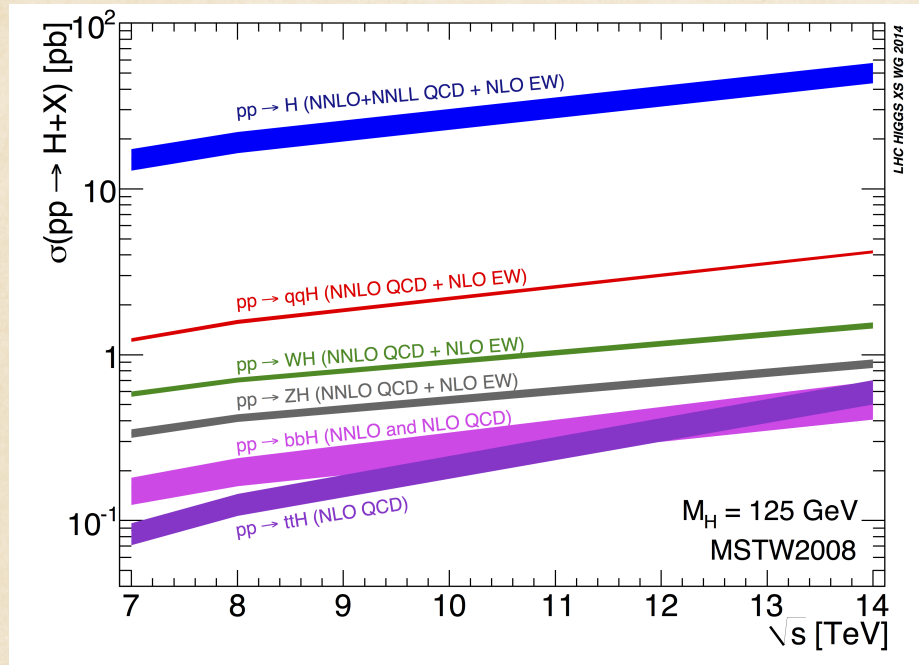


Figure 10.4: Higgs cross section as a function of center of mass energy of the collision. Part of the “band” is from uncertainty estimation via varying the renormalization (and factorization) scales.

2114 Real physics calculations often involve multiple dynamical scales, such
 2115 as relative momenta, particle masses, or kinematic variables. In these
 2116 cases, setting $\mu^2 = s$ may not be optimal, as it could introduce large
 2117 logarithms from other scales. Instead, we use physics intuition to select a
 2118 “best-guessed” central scale μ_0 :

- 2119 • Identify key scales (e.g., momentum transfer Q , jet p_T , masses m_q).
- 2120 • Common choices: Geometric means like $\sqrt{Q_1 Q_2}$ for two scales, or
 2121 dynamical options like $H_T/2$ (half the scalar sum of transverse
 2122 momenta) in multi-jet events.
- 2123 • In effective theories like SCET, multiple μ ’s handle different sectors
 2124 (hard, collinear, soft).

2125 To validate, compute at higher orders (e.g., NLO or NNLO), where μ
 2126 dependence should decrease as more terms cancel. Additionally, estimate
 2127 uncertainties via scale variation: Vary μ around μ_0 (typically by factors of
 2128 $1/2$ to 2), and independently vary related scales like the factorization
 2129 scale μ_F . The resulting band provides a conservative theoretical error
 2130 estimate. Stability under variation indicates reliability; otherwise, higher
 2131 orders or resummation may be needed.

2132 **This approach is iterative:** Start with a reasonable μ_0 , compute, vary
 2133 scales, compare orders, and refine based on the physics.

2134 IS THE “RENORMALIZATION GROUP” A GROUP?

2135 One might wonder if the “Renormalization Group” (RG) is, in fact, a
 2136 mathematical group. The name is a famous misnomer, and
 2137 understanding why it’s both wrong and right reveals the entire
 2138 conceptual leap from historical QFT to the modern, Wilsonian framework.

2139 THE CENTRAL QUESTION: A GROUP OR NOT?

2140 In the modern, arguably more general sense, the Renormalization Group
 2141 is **not a mathematical group**, but a **semigroup**.

2142 The confusion arises because the name is a historical artifact from an
 2143 older, more formal perspective. To see the difference, we must distinguish
 2144 between two different ways of thinking about the RG:

- 2145 1. The modern (Wilsonian) picture of a *physical flow*.
- 2146 2. The historical (QFT) picture of a *re-parameterization*.

2147 THE MODERN ANSWER: A SEMIGROUP OF PHYSICAL FLOWS

2148 When we think of the RG in the modern sense (thanks to Ken Wilson), we
 2149 are describing a physical process of “coarse-graining” or “integrating out”
 2150 high-energy (short-distance) degrees of freedom. We are literally
 2151 averaging over and discarding information.

2152 A mathematical **group** must satisfy four axioms:

- 2153 1. **Closure:** (Transform A) + (Transform B) = (New Transform C). (*This*
2154 *holds.*)
- 2155 2. **Associativity:** $(A+B)+C = A+(B+C)$. (*This holds.*)
- 2156 3. **Identity Element:** A transformation that does nothing. (*This*
2157 *holds.*)
- 2158 4. **Inverse Element:** For every transformation, there must be an
2159 inverse that perfectly undoes it. (***This fails.***)

2160 The Wilsonian RG fails on **Invertibility**.

2161 The act of integrating out high-energy modes is an **irreversible** process.
2162 You are averaging over details. Once you've coarse-grained a system, you
2163 have *lost information* about the precise, short-distance microscopic
2164 arrangement.

2165 You cannot run the RG “backwards” from a low-energy effective theory to
2166 uniquely discover the one, true high-energy UV theory. In fact, the entire
2167 concept of **universality** rests on this fact: many different UV theories can
2168 “flow” to the *same* low-energy IR theory.

2169 This is a **many-to-one mapping**. Because the transformations are not
2170 invertible, the RG is not a group. It is a semigroup (or more specifically, a
2171 monoid, as it has an identity element). This irreversibility is what gives
2172 the RG “flow” its famous “arrow of scale.”

2173 A DIFFERENT ASPECT: A LIE GROUP OF 2174 RE-PARAMETERIZATION

2175 So, why the name? The name is a historical artifact from the 1950s:
2176 Ernst Stueckelberg and André Petermann (1953) called it the “groupe de
2177 normalisation”, and Gell-Mann and Low (1954) developed the same idea
2178 independently for QED at short distances.⁴

2179 Their “group” was not about integrating out physical modes. It was about
2180 the **ambiguity** in the perturbative renormalization procedure. (This is
2181 how we introduced renormalization in the last part of Chapter II of our
2182 classes.)

- 2183 • In perturbative QFT, to get rid of infinities, we introduce an
2184 arbitrary, unphysical energy scale, μ , often called the
2185 “renormalization scale” or “subtraction point.”
- 2186 • All our renormalized couplings (g), masses (m), and field
2187 normalizations (Z) become functions of this arbitrary scale: $g(\mu)$,
2188 $m(\mu)$, $Z(\mu)$.
- 2189 • Physical observables (like a scattering cross-section) *cannot* depend
2190 on this arbitrary choice of μ .

⁴E. C. G. Stueckelberg and A. Petermann, *Helv. Phys. Acta* **26**, 499 (1953); M. Gell-Mann and F. E. Low, “Quantum electrodynamics at small distances,” *Phys. Rev.* **95**, 1300 (1954).

2191 • Therefore, a change in our choice of scale (from μ_1 to μ_2) must be
2192 perfectly compensated by a corresponding change in the couplings
2193 and fields to keep the physics invariant.

2194 This set of transformations is a group.

- 2195 • **Closure:** A transformation $\mu_1 \rightarrow \mu_2$ followed by $\mu_2 \rightarrow \mu_3$ is just the
2196 transformation $\mu_1 \rightarrow \mu_3$.
- 2197 • **Associativity:** This holds.
- 2198 • **Identity:** The transformation $\mu_1 \rightarrow \mu_1$.
- 2199 • **Inverse:** The transformation from $\mu_2 \rightarrow \mu_1$ is the *exact* inverse of
2200 $\mu_1 \rightarrow \mu_2$.

2201 This forms a continuous, invertible **Lie group**: a one-parameter group,
2202 the additive group of $\ln \mu$. The famous Callan–Symanzik and
2203 Gell-Mann–Low equations are the differential expressions (i.e., the Lie
2204 algebra) of this group. They describe how the couplings must “run” with
2205 scale μ to preserve the invariance of the physics.

2206 RECONCILIATION: IT IS UP TO THE PHYSICS CONTEXT

2207 One can rightly ask: “Even if I am given a single UV theory, like the
2208 Standard Model, its flow is described by a deterministic differential
2209 equation (the Callan-Symanzik equation). Can’t I just run this equation
2210 backwards and find the UV theory? Isn’t that an inverse?”

2211 That’s correct. If you are given **one single, well-defined theory**, the flow
2212 is just a **re-parameterization** of that one theory. You can slide your
2213 “viewing window” μ up and down, and the transformations are invertible.
2214 In this limited sense, for a *single theory*, the flow is a group.

2215 The Wilsonian revolution was a philosophical shift. We stopped thinking
2216 about a *single theory* (in particular, for condensed matter theorists) and
2217 started thinking about the **space of all possible theories** (so-called
2218 “theory space”).

2219 The physical act of coarse-graining is a *flow* in this giant space. Each
2220 point in the space is a different Lagrangian (a different set of all possible
2221 coupling constants).

2222 In this picture, we see universality. For example:

- 2223 • An Ising model on a simple cubic lattice.
- 2224 • An Ising model on a body-centered cubic lattice.
- 2225 • A real-world liquid–gas system near its critical point.

2226 These are three *wildly different* microscopic (UV) theories. But as we
2227 apply the Wilsonian RG (zoom out, coarse-grain, and rescale), they all
2228 flow toward the *exact same* point in theory space: the three-dimensional
2229 “Ising fixed point.”

2230 This is the **many-to-one** mapping that is the hallmark of a semigroup.
2231 The reason we prefer this picture is that it *explains the physics*. It
2232 explains *why* our low-energy world looks simple.
2233 The irreversible, information-losing, semigroup nature of the RG flow is
2234 what washes away all the complex, unknown details of Planck-scale
2235 physics, leaving us with the simple, renormalizable, effective field
2236 theories (like the Standard Model) that we use to describe our world.